

Multiple Choice

1. **Answer:** B

Options: A) Respiration; B) Photosynthesis; C) Bacterial decomposition; D) Combustion of fossil fuels

Note: Correct option: B - Photosynthesis

2. **Answer:** B

Options: A) Outer membrane, inner membrane, thylakoid membrane; B) Thylakoid membrane, inner membrane, outer membrane; C) Inner membrane, thylakoid membrane, outer membrane; D) Thylakoid membrane, outer membrane, inner membrane; E) Strama, outer membrane, inner membrane

Note: Correct option: B - Thylakoid membrane, inner membrane, outer membrane

3. **Answer:** D

Options: A) All phenotypic variation is the result of genotypic variation.; B) All genetic variation produces phenotypic variation.; C) All nucleotide variability results in neutral variation.; D) All new alleles are the result of nucleotide variability.

Note: Correct option: D - All new alleles are the result of nucleotide variability.

4. **Answer:** A

Options: A) Smaller cells are more resilient to external damage.; B) Smaller cells have a larger number of organelles.; C) Smaller cells require less energy to function.; D) Smaller cells can multiply faster.; E) Smaller cells fit together more tightly.

Note: Correct option: A - Smaller cells are more resilient to external damage.

5. **Answer:** D

Options: A) Promoter; B) Inducer; C) Repressor; D) Operator; E) Inhibitor

Note: Correct option: D - Operator

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Estrogen'.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '1 / 36'.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Sister chromatids separate during meiosis I.'

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Through operant conditioning'.

Long Form Response

11. **Answer:** The bacterial cell can live without its cell wall in an isotonic medium, where the osmotic pressure is equal to the bacterial protoplasm. The cell wall of bacteria is composed of insoluble peptidoglycan, while the cell wall in eucaryotes is composed of cellulose.. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** The concentration of inorganic salts has no effect on blood pressure.. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key